WEEK 10 QUESTION BANK

1. Create a complete end to end Neural network for MNIST(classify handwritten numerals)
2. Build a shallow Neural network with Keras Dense Layers.
3. A student wants to implement a simple binary classification problem using a Perceptron. Explain the working of a Perceptron with input values, weights, bias and activation function.
4. An organization is choosing between Machine Learning and Deep Learning for a project involving a large dataset. Differentiate between Machine Learning and Deep Learning based on dataset size, hardware requirements and training time.
5. A developer wants to overcome the dying ReLU problem in a neural network. Identify the suitable activation function and explain how it solves the problem.
6. A neural network produces incorrect predictions during training. Explain how forward propagation and backward propagation help in improving the predictions.
7. Consider a neural network that receives three input values. Explain how weights and bias are used to calculate the output of an artificial neuron.
8. A developer wants to design a neural network that produces output values between 0 and 1. Identify the suitable activation function and explain its working.
9. A machine learning model predicts the price of a product as ₹550, whereas the actual price is ₹500. Calculate the squared loss using the cost function and explain the result.
10. Explain why TensorFlow is used in machine learning and discuss any four usecases/applications of TensorFlow.
11. Explain the three programming models available in Keras with suitable examples.